Azure VM Cheat Sheet (Azure CLI)

1. Sign in & set up

Goal	Command
Log in	`az login`
List your subscriptions	`az account list -o table`
Pick a subscription	`az account set --subscription "<name-or-id>"`
Create a resource group	`az group create -n <rg> -l <region>`
List your resource groups	`az group list -o table`

2. Find an image

Goal	Command
List common image **aliases** (e.g. `Ubuntu2404`)	`az vm image list -o table`
List image **publishers**	`az vm image list-publishers -l <region> -o table`
List **offers** from a publisher	`az vm image list-offers --publisher Canonical -l <region> -o table`
List **skus** for an offer	`az vm image list-skus --publisher Canonical --offer ubuntu-24_04-lts -l <region> -o table`
List actual image **versions**	`az vm image list --publisher Canonical --offer ubuntu-24_04-lts --sku server --all -l <region> -o table`

An image is identified by a 4-part **URN**: `Publisher:Offer:Sku:Version` (e.g. `Canonical:ubuntu-24_04-lts:server:latest`). For common images you can use a short alias instead (e.g. `Ubuntu2404`).

3. Find a VM size

Goal	Command
List sizes in a region	`az vm list-skus -l <region> --resource-type virtualMachines -o table`
Sizes with vCPUs + RAM (filtered)	`az vm list-skus -l <region> --resource-type virtualMachines --query "[?starts_with(name, 'Standard_B2')].{name:name, vCPUs:(capabilities[?name=='vCPUs'].value | [0]), memGB:(capabilities[?name=='MemoryGB'].value | [0])}" -o table`
What can I **resize** this VM to?	`az vm list-vm-resize-options -g <rg> -n <vm> --query "[].name" -o table`

4. Create the VM

Goal	Command
Create (SSH-key auth)	`az vm create -g <rg> -n <vm> --image Ubuntu2404 --size Standard_B2als_v2 --admin-username azureuser --generate-ssh-keys`

Goal	Command
Create (password auth)	`az vm create -g <rg> -n <vm> --image Ubuntu2404 --size Standard_B2als_v2 --admin-username azureuser --admin-password '<StrongP@ssw0rd!>'`
Create + run a startup script (cloud-init)	`az vm create -g <rg> -n <vm> --image Ubuntu2404 --size Standard_B2als_v2 --admin-username azureuser --generate-ssh-keys --custom-data cloud-init.yaml`
See every option	`az vm create --help`

5. Inspect, IP, connect

Goal	Command
Show the full VM definition	`az vm show -g <rg> -n <vm> -o jsonc`
Check the power state	`az vm get-instance-view -g <rg> -n <vm> --query "instanceView.statuses[?starts_with(code,'PowerState/')].displayStatus" -o tsv`
Get the **public IP**	`az vm list-ip-addresses -g <rg> -n <vm> -o table`
Public IP into a variable	`IP=$(az vm list-ip-addresses -g <rg> -n <vm> --query "[0].virtualMachine.network.publicIpAddresses[0].ipAddress" -o tsv)`
Connect (SSH)	`ssh azureuser@<public-ip>`
Connect using a specific key	`ssh -i ~/.ssh/id_rsa azureuser@<public-ip>`
Open a port (e.g. 80)	`az vm open-port -g <rg> -n <vm> --port 80 --priority 900`

6. Manage power, size, lifecycle

Goal	Command
Start	`az vm start -g <rg> -n <vm>`
Stop the OS (still billed for compute)	`az vm stop -g <rg> -n <vm>`
Deallocate (stops compute billing)	`az vm deallocate -g <rg> -n <vm>`
Restart	`az vm restart -g <rg> -n <vm>`
Resize	`az vm resize -g <rg> -n <vm> --size Standard_B4als_v2`
List all your VMs	`az vm list -d -o table`

7. Clean up (stop paying)

Goal	Command
Delete just the VM	`az vm delete -g <rg> -n <vm> --yes`
Delete everything (VM + disk + NIC + IP + NSG)	`az group delete -n <rg> --yes --no-wait`

Cost tip: "Stop" in the OS or `az vm stop` still bills you for compute. Use **`az vm deallocate`** to pause billing while keeping the VM, and `az group delete` to remove everything when you're done.